ACCIOJOB • FINAL PROJECT PRESENTATION

OLYMPIC GAMES DATA ANALYSIS

Historical trends in sports, athletes, medals and regional representation

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PostgreSQL • Python • Pandas • Excel • Power BI

Objective and problem statement

Why this analysis was undertaken

Objective: turn historical Olympic records into clear, comparable insights about Games editions, sports, athletes, medals and regions.

The challenge

The data is spread across 11 related tables and multiple record types.

The approach

Use SQL to connect and aggregate records, then organize results and visualize patterns.

The outcome

18 EDA result sheets, project documentation and a five-page Power BI report.

The project is descriptive: it identifies patterns in the supplied dataset and does not claim to prove historical causes.

Data dictionary: 11-table model

Entities and bridge tables in the PostgreSQL database

Games & locations	**games, city, games_city**	Olympic editions, seasons and host-city associations
Athletes & entries	**person, games_competitor**	Athlete attributes and participation by Games edition
Sports & events	**sport, event**	Sport categories and individual Olympic events
Medals	**medal, competitor_event**	Medal categories and competitor-event outcomes
Regional mapping	**noc_region, person_region**	NOC / region labels and person-region associations

Key modeling idea: bridge tables connect entities; participation rows, people and medal records are different units.

Tools and methodology

How the analysis was carried out

SQL / PostgreSQL	Query relational tables and calculate grouped results.
SQLAlchemy	Create the Python connection and execute SQL queries.
Pandas	Receive query outputs as DataFrames for inspection and organized handling.
Excel	Keep the 18 query outputs and supporting sheets together.
Power BI	Model relationships and present results across five report pages.

Data path: PostgreSQL → SQLAlchemy/Python → Pandas DataFrames → Excel and Power BI.

Finding 1: the Games expanded in scale

Counts reported by the EDA analysis

Summer events

42 (1896) → 306 (2016)

Summer participants

72 (1896) → 11,176 (2016)

Winter events

17 (1924) → 98 (2014)

The recorded dataset shows substantial growth in both Summer and Winter competition scale.

- Summer participant entries reached 11,176 in the 2016 edition in the query results.
- Winter participant entries increased from 265 in 1924 to 2,743 in 2014.
- These are dataset counts; they should be described using the same definitions as the source queries.

Finding 2: sports and event mix vary

Event counts and name-based gender categories

Athletics

Among the highest event counts in the displayed analysis.

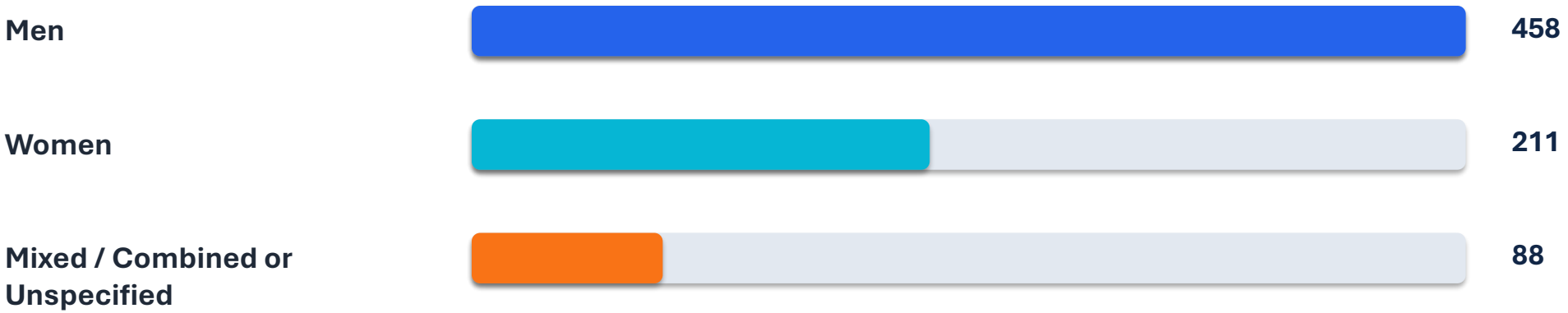
Shooting

Also among the sports with the highest event counts.

Programme changes

First/last observed years identify cases for follow-up.

Event classification in the EDA results



Categories depend on event-name classification; they are not a complete historical measure of gender equity.

Finding 3: participation and athlete demographics

Use regional and age comparisons with context

Regional participation

USA is the largest region in the visible participant chart, followed by the UK and France.

Gender entries

The dashboard shows roughly 132K male and 48K female participation entries.

Average age

The displayed edition-level averages are generally in the low-to-upper twenties.

Interpretation notes

- Participation entries do not necessarily equal unique individuals.
- Height and weight analyses should account for zero or placeholder values.
- Changes in the sports represented can affect average age and body-measurement patterns.

Finding 4: medal records differ by sport and NOC

Based on the medal-analysis visuals and EDA results

Gold records

USA ranks first in the displayed gold-medal-by-NOC chart.

Sports

Athletics has the highest total medal-record count among the visible sports.

Repeat presence

UK: 29 Summer editions; Canada: 22 Winter editions with medal presence in the EDA.

Counting caveat

- Competitor-event records may count multiple athletes from a team event.
- Medal-bearing records should not be described as unique physical medals unless deduplicated.
- Regional totals need historical context because NOC and region labels change over time.

Power BI report: five-page design

Use the existing dashboard screenshots in the live walkthrough

1 • Olympic Overview

Editions by season, city and decade

2 • Sports & Events

Sport counts, gender categories and events by edition

3 • Athlete Demographics

Participants by region, gender entries and average age

4 • Medal Analysis

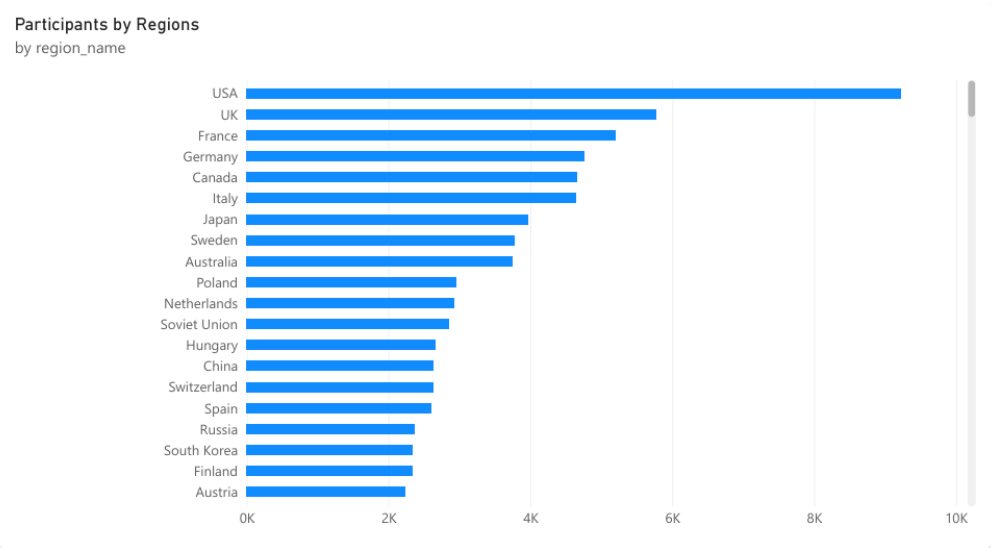
Medal records by edition, NOC and sport

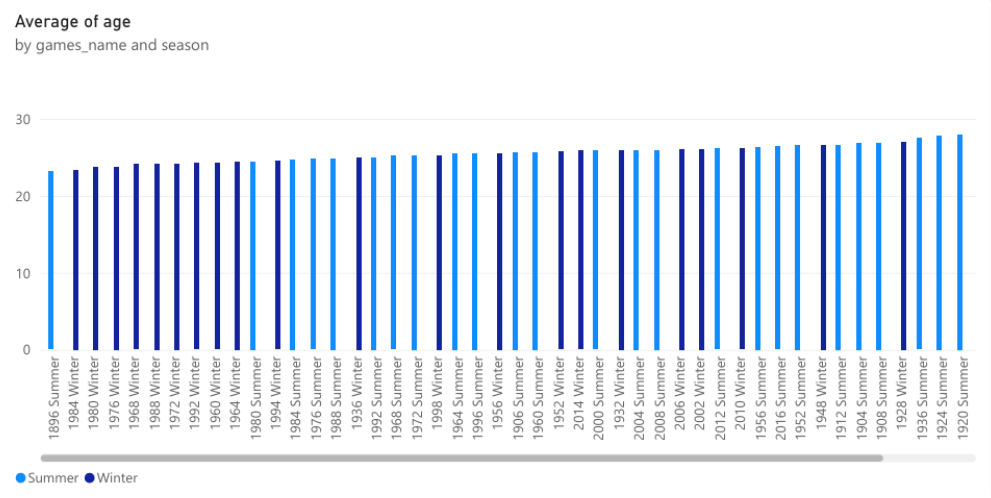
5 • Regional Participation

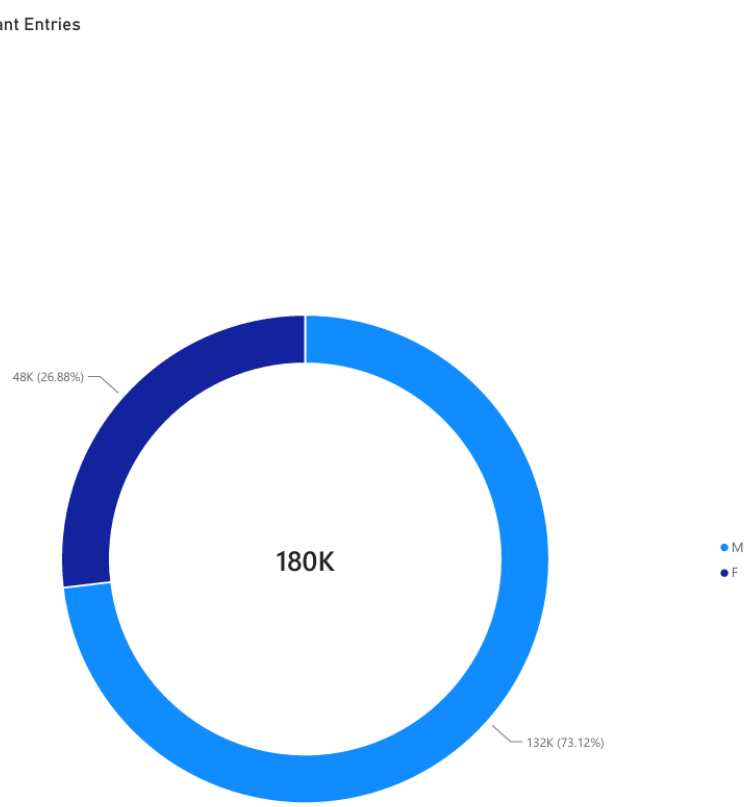
Regions represented over time and medal categories

The dashboard groups related questions into pages instead of giving every EDA question a separate slide.

Sample Output Using PowerBI







Recommendations

Practical next steps supported by the analysis

- Track participation and event counts across future Olympic editions to monitor programme growth.
- Validate first-observed and last-observed event years against official programme histories before labeling events as introduced or discontinued.
- Compare medal performance within individual sports, not only by overall regional totals.
- Document data definitions for unique athletes, participation entries and medal-bearing records.
- Review placeholder measurements, blank medal labels and historical region mappings.
- Update the Power BI report when new editions or improved data become available.

Recommendations are analytical and reporting improvements, not causal claims about Olympic policy.

Limitations and conclusion

What the project establishes

Limitations

Dataset ends with 2016 Summer and 2014 Winter editions; regional labels, event classifications and measurement completeness require context.

Conclusion

SQL, Pandas, Excel and Power BI were combined to organize and communicate 18 exploratory analyses across five dashboard pages.

Takeaway

The project turns a relational historical dataset into structured, reviewable and visual insights—while keeping the limits of the underlying records explicit.

Thank you

Project deliverables and supporting materials

For review and reproducibility

Excel workbook

18 EDA result sheets plus summary and supporting sheets.

Word report

Database structure, methodology, findings, limitations and recommendations.

Power BI report

Five pages covering overview, sports, demographics, medals and regions.

Tools and references

- PostgreSQL documentation: postgresql.org/docs/
- Pandas documentation: pandas.pydata.org/docs/
- SQLAlchemy documentation: docs.sqlalchemy.org/
- Power BI documentation: learn.microsoft.com/power-bi/